Vishek Masalia

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Education

Johns Hopkins University (JHU), Baltimore, MD Expected: May 2027

Master of Science in Engineering in Robotics GPA: 3.89/4.0

- ***Coursework:*** *Robot Dynamics & Kinematics, Motion Planning, Mechatronics, Haptic Interface Design, Machine Learning, Deep Learning, Algorithms for Sensor-Based Robotics*

University of Illinois Chicago (UIC), Chicago, IL Aug 2021 - Dec 2024

Bachelor of Science in Mechanical Engineering GPA: 3.94/4.0

- ***Activities:*** *ASME, SAE (Formula SAE Team)*
- ***Coursework:*** *Mechanical Design, Mechatronics, Heat Transfer, Numerical Methods*

Technical Skills

- **Robotics & Control:** ROS 2, MoveIt, Rviz, MATLAB RTDE, Robot Kinematics & Dynamics, Motion Planning, Cartesian Control (IK, Resolved-Rate), Closed-Loop Control
- **Programming:** C++, C (Embedded), Python, MATLAB, Linux (Ubuntu)
- **Mechanical & Mechatronic Design:** System Design, DFM/DFA, Rapid Prototyping, Failure Analysis, Test & Validation
- **CAD & Simulation:** SolidWorks (CSWA), Onshape, ANSYS (FEA), Tolerance Analysis
- **Fabrication:** 3D printing, Machining, Laser Cutting, Mechanical Assembly, Wood Shop

Projects

UR5e Autonomous Push-and-Place Manipulation - JHU, Baltimore, MD Nov 2025 – Dec 2025

Robotics Control Engineer

- Developed autonomous push-and-place and near-contact manipulation tasks on a UR5e industrial robot
- Implemented and benchmarked Inverse Kinematics (IK) and Resolved-Rate (RR) Cartesian control for trajectory execution
- Designed IK solution filtering to enforce joint feasibility, continuity, and avoidance of large joint-space discontinuities
- Built control pipelines in ROS 2 (simulation + hardware) and MATLAB RTDE, validating performance across platforms
- Planned motion in SE (3) and taught initial configurations using free drive mode
- Achieved highly repeatable straight-line Cartesian motion and precise near-contact positioning without object deformation
- Demonstrated lower latency and higher reliability in ROS-based execution compared to RTDE

Haptic Grasping Feedback for Surgical Robotics - JHU, Baltimore, MD Aug 2025 – Dec 2025

Project Lead – Mechatronic Systems and Haptics

- Led development of a kinesthetic haptic interface to restore force feedback in da Vinci-style surgical grippers
- Designed and fabricated a capstan-driven actuation system using a brushed DC motor for force rendering
- Integrated position sensing (MR sensors) and force sensing (FSRs) for closed-loop interaction
- Developed real-time embedded C++ control architecture on Hapkit platform
- Implemented physically based interaction models (spring, damping, friction, impact) for tissue simulation
- Applied signal filtering and state estimation to ensure stable, noise-robust force output
- Enforced safety constraints (≤ 5 N force limit) in hardware and software
- Successfully rendered distinct interaction states: free space (0 N), soft tissue (0–2 N), hard tissue (1–4 N)
- Identified motor commutation as source of force ripple, proposing brushless upgrade for improved fidelity

CLARIFEYE: Low-Cost Fundus Imaging System- JHU, Baltimore, MD Aug 2025 – Dec 2025

Hardware Design Lead

- Co-founded a startup developing AI-assisted retinal screening for low-resource settings
- Designed a smartphone-agnostic handheld fundus imaging attachment using a 20D lens system
- Engineered X–Y alignment and axial (Z-axis) focusing mechanisms for consistent optical alignment across devices
- Built a fully mechanical, modular system enabling rapid deployment (<4 min screening) at less than $100 cost
- Developed complete CAD architecture in SolidWorks with DFM-focused modular components
- Optimized ergonomics and structural stability to reduce motion artifacts and improve image quality
- Ensured compatibility with downstream AI pipelines (enhancement, stitching, GAN restoration, classification)
- Positioned system as a scalable alternative to high-cost clinical imaging devices

Projects (Continued)

Adaptable Pedal Box - UIC Formula SAE Team, Chicago, IL Aug 2023 – May 2024

Mechanical Design & Prototyping Engineer

- Designed a pedal box system compatible with both electric and combustion racecars
- Developed full CAD assemblies in SolidWorks with emphasis on packaging, ergonomics, and manufacturability
- Performed FEA in ANSYS to validate structural integrity under operational loads
- Iterated designs based on simulation and testing, producing functional prototypes via laser cutting and 3D printing
- Delivered final validated prototype meeting all FSAE regulations and integration constraints
- Presented at UIC Engineering Expo, awarded "Best in Show"

Autonomous Writing Device - UIC, Chicago, IL Aug 2021 – Dec 2021

Mechanical & Embedded Systems Developer

- Designed and built a fully functional automated writing system using Arduino UNO
- Developed mechanical structure using MDF, 3D-printed components, and rack-and-pinion actuation
- Programmed control logic in C++ for precise motion and repeatable drawing
- Demonstrated reliable performance in drawing shapes and letters with minimal vibration

Robotics Competitions - FTC, WRO, FIRST Programs, India Jun 2017 – May 2020

Project Lead (2020), Design Lead (2017-2020)

- Designed, built, and programmed autonomous and teleoperated robots for national-level competitions
- Led mechanical design and system integration using SolidWorks, 3D printing, and modular hardware systems
- Developed control algorithms for autonomous navigation (line following, task sequencing) and manipulation
- Executed iterative prototyping cycles under strict competition constraints (time, robustness, scoring objectives)

Work Experience

Engineering Success Program - UIC, Chicago, IL Aug 2024 – Dec 2024

Teaching Assistant

- Led ENGR 100 sessions for 200+ students, covering foundational engineering concepts
- Mentored freshman and transfer students in adapting to engineering coursework and problem-solving approaches
- Guided students through a design project applying heat transfer and thermodynamics principles
- Collaborated with 30+ teaching staff to improve student engagement and learning outcomes

Anand Research Group - UIC, Chicago, IL Jun 2022 – Aug 2022

Research Intern

- Conducted experimental studies on ice adhesion strength across metallic surfaces with varying salt solutions
- Performed material characterization of polymers for cold-environment applications
- Analyzed and visualized experimental data using Excel for comparative evaluation
- Contributed to research on phase change materials (PCM) for frost mitigation
- Acknowledged in publication: *"How to Select Phase Change Materials for Tuning Condensation and Frosting"*

Achievements

- **3rd Place** – WRO (2019), India
- **Collins Aerospace Innovative Award (Most Creative Robot Design) –** FTC (2019-20), India
- **Finalist Alliance 1st Runner-Up** – FTC (2017–18), India